

duction development by technology, by 2011, thin film had accounted for 3204 MWp, ribbon-Si 120 MWp, multi-Si 10,336 MWp and mono-Si 9114 MWp, respectively.

Efficiencies in the solar cell market

Efficiencies in the solar cell market range from 1 to 5 per cent for organic, dye and nanostructure cells. PV technologies of interest in the next one to two decades are:

1. 6-11 per cent: Thin-film cells (a-Si, microcrystalline-Si, CIS, CIGS, CdTe)
2. 14-18 per cent: mc-Si, umg-Si, simple c-Si cells
3. 20-24 per cent: High-efficiency c-Si cells
4. 36-41.1 per cent: High-efficiency III/V tandem cells for concentrators with 25-30 per cent module efficiency

The price learning curve for all c-Si PV technologies indicated that with each doubling of cumulative production, price went down by 20 per cent. So thin-film technologies must ramp up fast enough to maintain a clear cost advantage at lower efficiencies!

Prof. Weber cited examples: Solar cells built with 100 per cent umg-Si have a conversion efficiency of less than 20 per cent. Silicor plans a umg-Si plant with a capacity of 16,000 tpa at a cap-ex of \$600 million. The median efficiency at CaliSolar is now 16.6 per cent. At least 11 per cent of the cells have a conversion efficiency of above 17 per cent with the highest at 17.7 per cent. Q-cells has achieved 18.2 per cent efficiency using umg-Si cell with backside contacts.

The advantages and future requirements for mc-Si include a mature process and no scalability limit. Quality needs to be high enough for 20 per cent efficient solar cells. Diffusion length should be greater than 500 μm at cell thickness of 150 μm . Impurities should exhibit low activity/good gettering behaviour. There should be monocrystals, leading to easy texturing. Some other requirements include increase of yield (less low-quality areas), processes

Creating market barriers will lead to higher prices and less pressure for cost reduction and innovation, ultimately hurting the economies that adopt them

for umg-Si feedstock and cost below 0.30/Wp.

Prof. Weber gave another example of a high-quality block-crystallised silicon material. There should be polarity switch in umg silicon. The umg Si is compensated: both boron (B) and phosphorous (P) are present in the feedstock. The dopant crossover is due to different segregation coefficients. The consequence: p- and n-type Si within the same brick and even single wafers. Dopant engineering is needed to avoid the p-n switch or increase the yield.

There is another example: non-conventional c-Si material or solar cells made from crystalline silicon thin-films. All concepts have good to very good cost perspective. High-throughput, low-cost Si deposition will be required for quick progress.

The key design data of a non-conventional c-Si material (ProConCVD) was presented. The ProConCVD is a massively scaled version of the ConCVD, intended to prove the scalability of the approach to a near-production level of more than a thousand wafers per hour. The data includes:

1. Three tracks, each with two carriers. Each carrier holds three $156 \times 156 \text{ mm}^2$ wafers in height
2. Total deposition area: 5 m^2
3. Maximum transport speed: 12 m/h
4. Furnace: max. 360kW, resistance-heated, $2 \times 8 \text{ m}^2$ footprint, 2m stable zone
5. Process temperature up to 1300°C
6. Available process gases (maximum consumption/min): SiHCl_3 (300 gm), SiCl_4 (500 gm), $\text{SiCl}_3(\text{CH}_3)$ (300 gm), H_2 (4000 sl), HCl (50 sl)



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